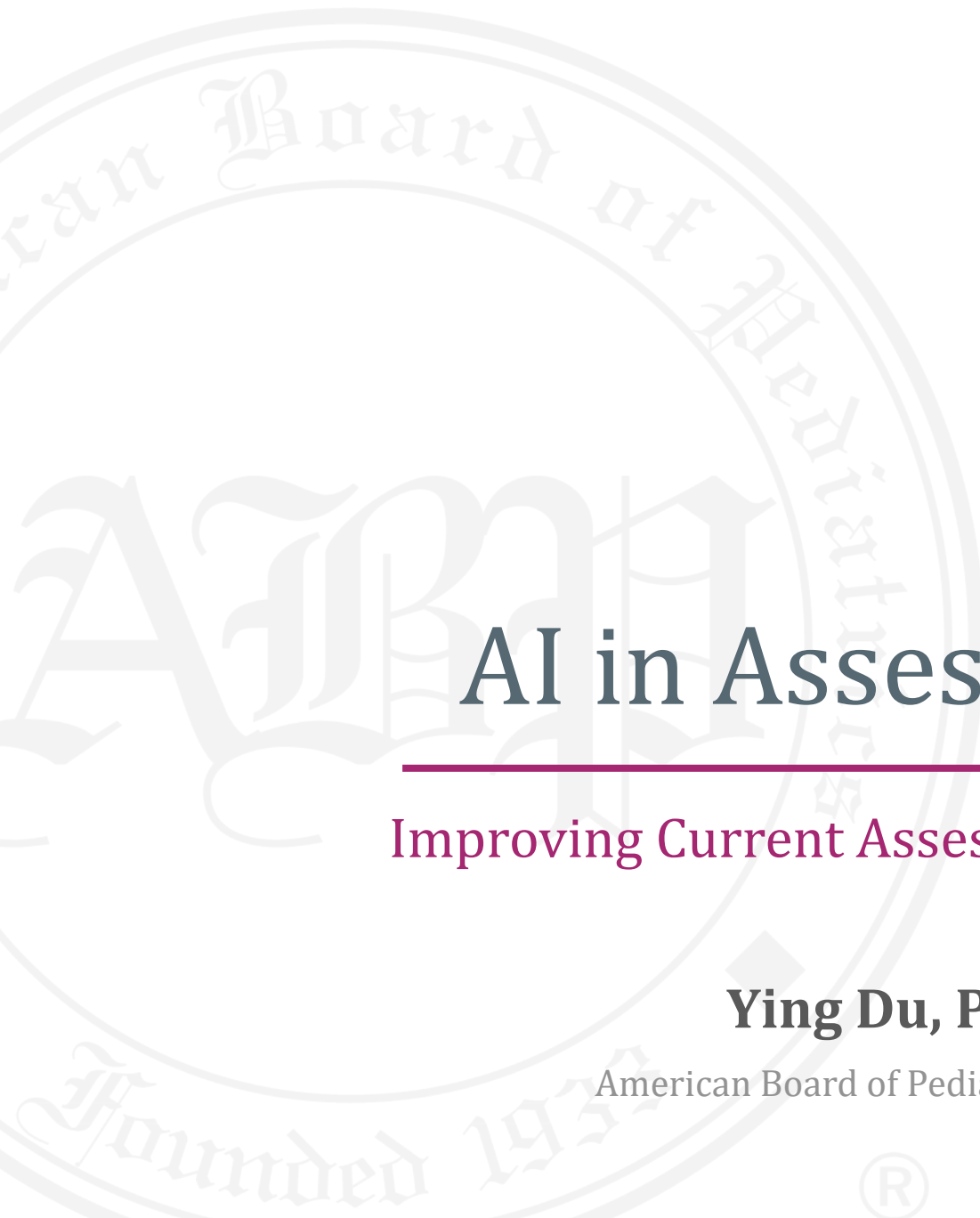




AI in Assessment at the ABP

Improving Current Assessment • Developing New Assessment

Ying Du, PhD • Ting Wang, PhD

American Board of Pediatrics | ABMS Conference, September 2026

Disclosure

- Nothing to Disclose

Two Complementary Initiatives

Using AI to improve efficiency in the ABP's existing assessments — and measuring a new construct in a new assessment product.

EXISTING ASSESSMENTS

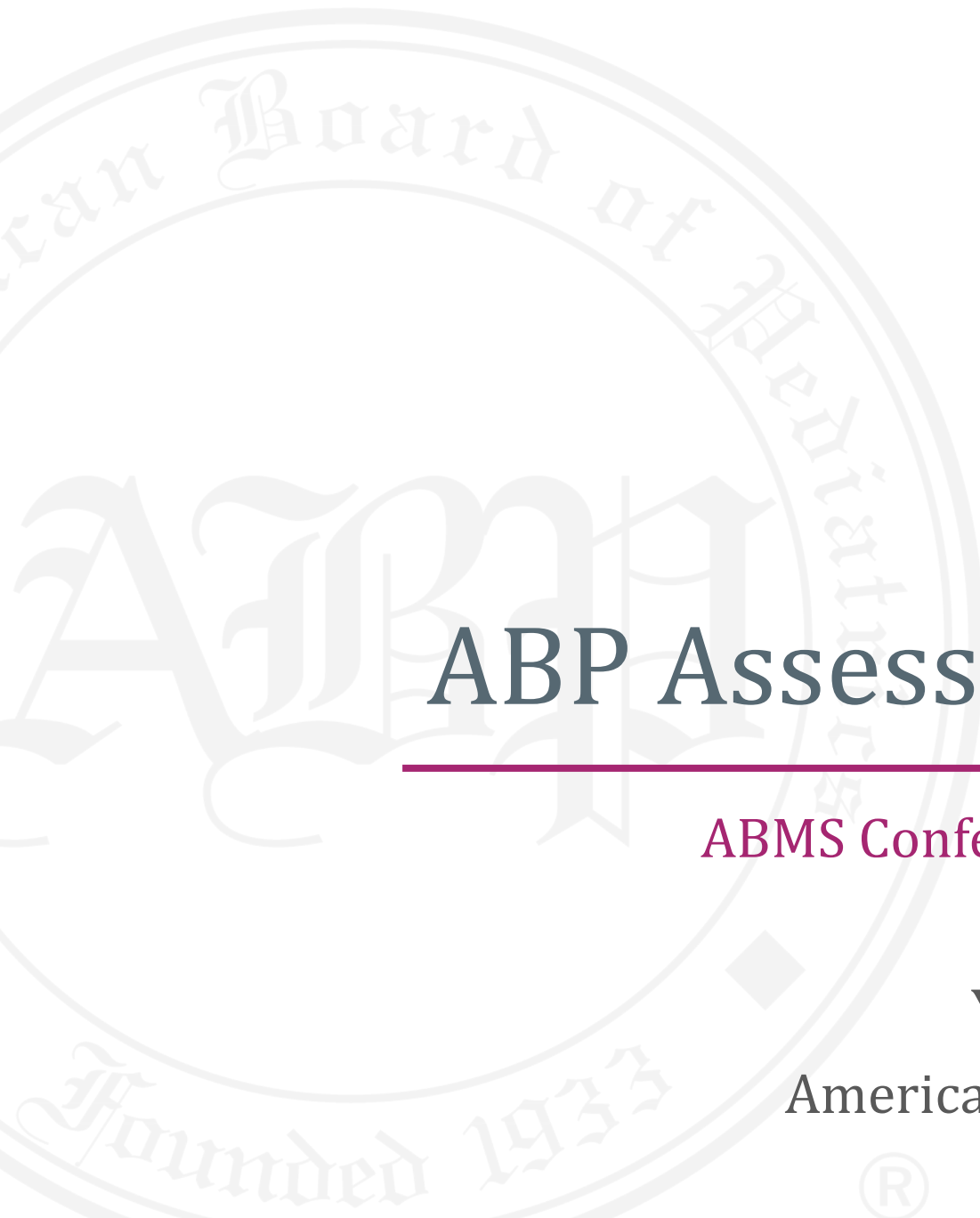
Ying Du, PhD

- **ABP Assessment AI Roadmap**
- AI to improve efficiency of current products
- Governance and success metrics

NEW ASSESSMENTS

Ting Wang, PhD

- **Measuring Clinical Reasoning Under Uncertainty**
- A new item type for a new construct
- Validity evidence



ABP Assessment AI Roadmap

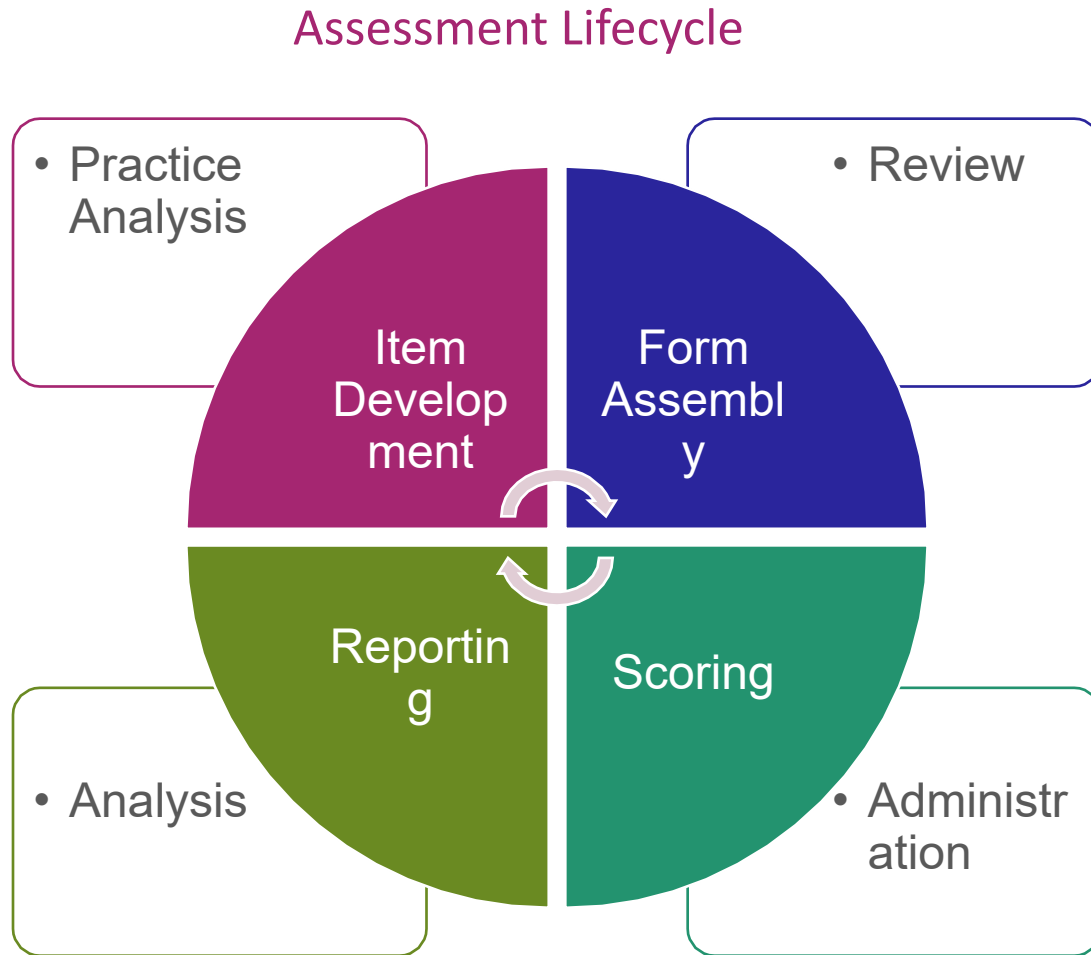
ABMS Conference | September 2026

Ying Du, Ph.D.

American Board of Pediatrics

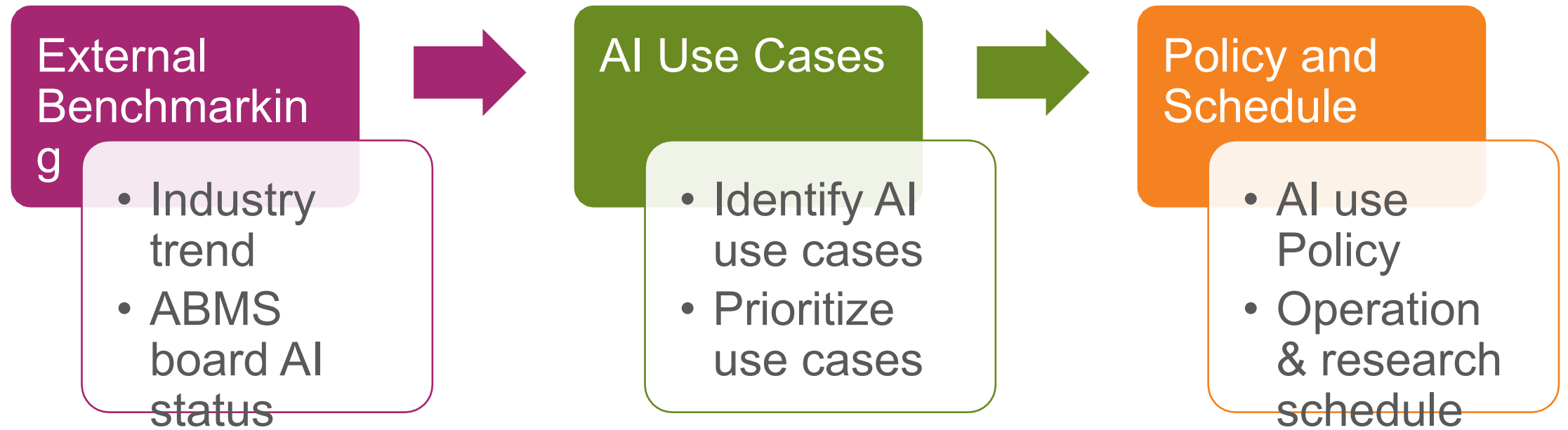


Problems in Assessment Lifecycle



- Manual work
- Repetitive review
- SME burden
- Complex QC
- Staff time
- Scalability

How we build the roadmap



Where is the field today?



AI Adoption Trends

- AI adoption is moving from experimentation to operational use.
- AI is increasingly used across the assessment lifecycle

Major Application Areas

- Blueprint automation and MCQ item development
- Stronger promise for item classification and workflow efficiency than for writing full items.

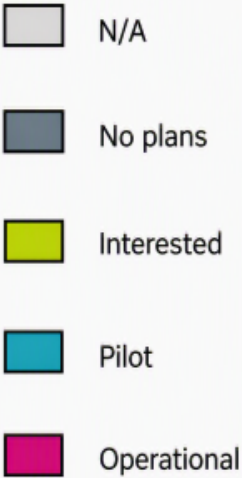
Cautious Use in High-Stakes Testing

- Restraint in using AI for core psychometric functions
- Support tool rather than a replacement for psychometric expertise

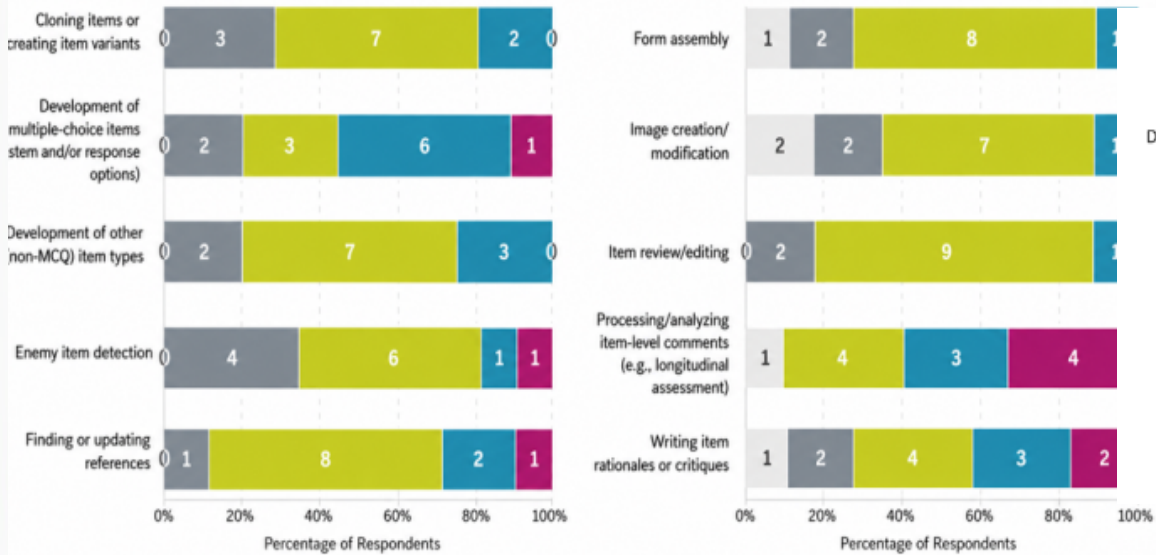
Major Barriers in AI Application

- Lack clear candidate AI-use policies
- Concerns about copyright, cost, and maintaining exam integrity

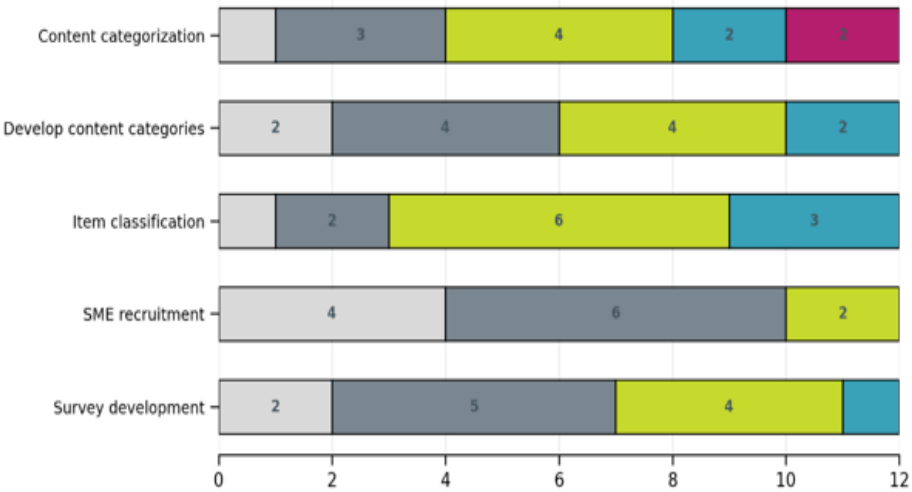
ABMS Survey Results



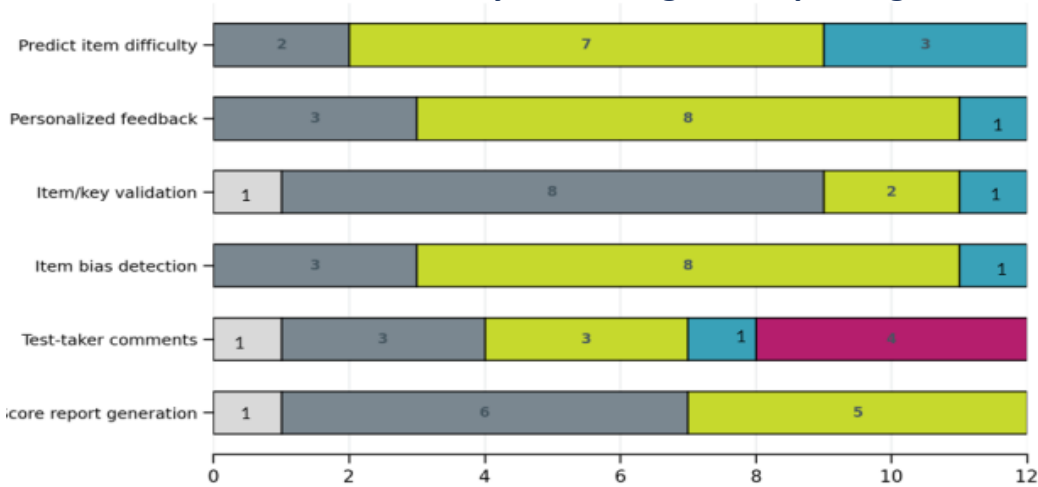
AI Use in Item Development Tasks (Count of Respondents)



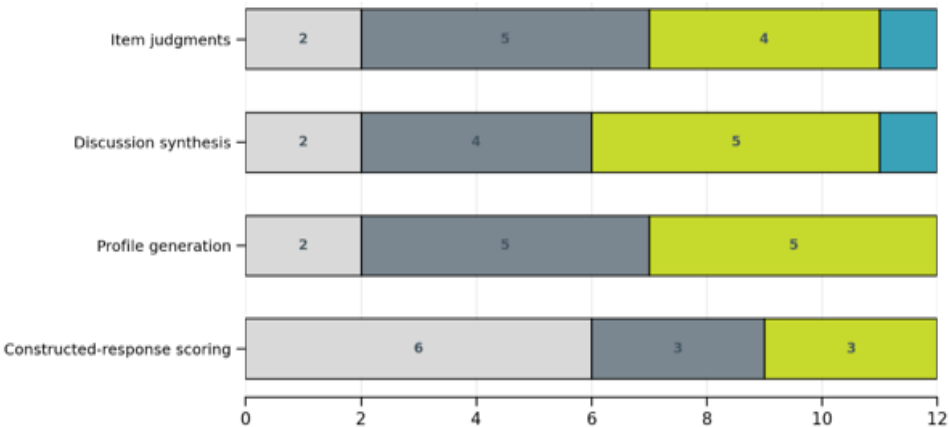
Practice Analysis



Analysis, Scoring and Reporting



Standard Setting



ABMS Survey Summary



AI adoption across ABMS boards is widespread, but still early and cautious, mostly exploratory.



AI is primarily used for lower-risk support functions, focusing mostly on efficiency rather than decision-making.



Governance is developing faster than operational implementation, and vendor platforms are driving some AI exposure and adoption.



Boards remain cautious in high-stakes psychometric activities.



Dedicated AI staffing remains limited, and task-specific training is lacking.

Framework

A staged adoption model:

- lower-risk, efficiency-oriented AI applications first;
- core assessment decisions later, only if validated.

Opportunity

- Reduce repetitive workload
- Improve consistency in selected workflows
- Support faster content development and qualitative

Boundary

- No autonomous certification decisions
- No use of unapproved tools with secure content
- Human review and documentation required

Identify AI Use Cases

Start with the pain points, not technology



Test
Development



Practice
Analysis



Psychometrics
Analysis and
Scoring



Test Security



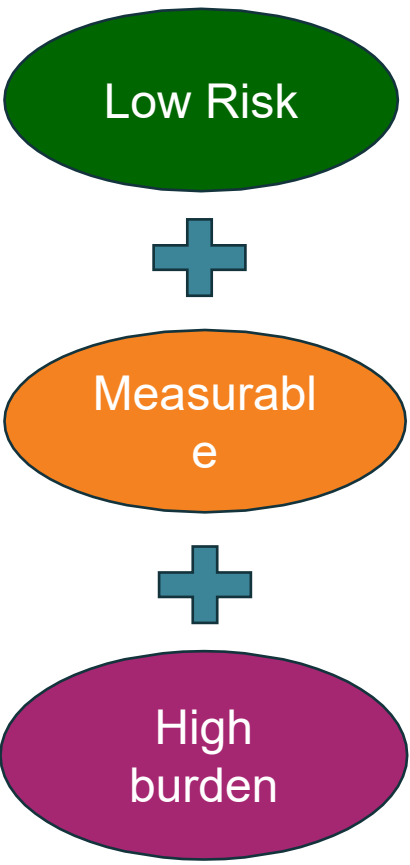
Content Quality
and
Documentation



Candidate and
Program
Reporting

AI Use Case Evaluation

Domain	Description	Weight
Strategic Value & Impact	- Improve exam validity or reliability? - Reduce the "pain" of repetitive data tasks?	50%
Organizational Readiness	- Data already clean and accessible? - Internal skills (or a vendor partner) to implement?	30%
Cost & Risk	- AI "hallucinates" a result? - Potential bias introduced into assessments? - Long-term maintenance cost?	20%



Phased, Evidence-Based Adoption

Phase 1 Research & Operational Efficiency

- Research, pilot testing, and low-risk applications
- Parallel evaluation of AI vs. traditional processes
- No changes to high-stakes decisions
- Build internal capability and evidence

Phase 2 Operational Integration & Validation

- Controlled operational integration of validated use cases
- Human-in-the-loop workflows
- Ongoing performance and risk monitoring
- Psychometric validation reports

AI Integration Roadmap (Phase 1 & Phase 2)

2026 – 2029 Timeline

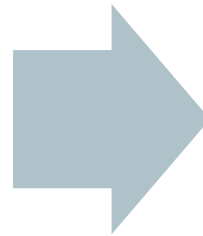
Project	Phase	Owner	2026				2027				2028				2029			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Project #1	Phase 1	Test Development																
Project #2	Phase 1	Test Development																
Project #3	Phase 1	Test Development																
Project #4	Phase 1	Test Development																
Project #5	Phase 1	Psychometrics																
Project #6	Phase 1	Psychometrics																
Project #7	Phase 1	Psychometrics																
Project #8	Phase 1	Psychometrics																
Project #9	Phase 1	Psychometrics																
Project #10	Phase 1	Psychometrics																
Project #1	Phase 2	Test Development																
Project #2	Phase 2	Test Development																
Project #3	Phase 2	Test Development																
Project #4	Phase 2	Psychometrics																
Project #5	Phase 2	Psychometrics																
Project #6	Phase 2	Test Development & Psychometrics																

Success Metrics

HOW WE WILL KNOW AI ADOPTION IS WORKING

Phase 1

- * High-quality research completed
- * AI vs. human comparability
- * Model accuracy & consistency
- * SME satisfaction & time savings
- * Operational feasibility



Phase 2

- * Psychometric quality maintained
- * Workflow adoption
- * Operational efficiency gains
- * Governance & monitoring
- * Stakeholder acceptance

Evidence-based progression to operational AI only after quality and human oversight are demonstrated.

Key Risks and Mitigation

RESPONSIBLE ADOPTION REQUIRES ACTIVE MANAGEMENT

Validity / quality risk

Structured evaluation metrics; AI-human comparisons; SME review

Automation bias

Human approval; staff and SME training to scrutinize outputs

Security risk

Approved environments only; no secure content in public tools

Capability gaps

AI literacy, prompting skills, and review protocols

Resource constraints

Prioritize by value, readiness, and risk

Stakeholder trust

Document evidence, governance, monitoring, and boundaries

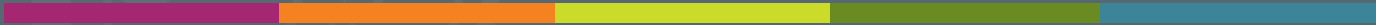
Lessons Learned

- ✓ Start with the problem, not the AI
- ✓ Separate experimentation from operationalization
- ✓ Build the evidence and governance infrastructure alongside the technology



Successful AI Integration = Efficiency + Quality + Adoption + Trust

Thank you!



Contact us at ydu@abpediatrics.org

The **American Board of Pediatrics (ABP)** was founded in 1933. It is one of the 24 certifying boards of the American Board of Medical Specialties (ABMS). The ABP is an independent and nonprofit organization. The ABP's mission is to advance child health by certifying pediatricians who meet standards of excellence and are committed to continuous learning and improvement.

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Measuring Clinical Reasoning Under Uncertainty

New Item Type

Ting Wang, PhD
Director of Assessment Innovation Research

What Multiple Choice Measures — and What It Misses

▪ MCQs measure well

- Factual knowledge and recall
- Application in clearly-defined situations
- Binary right-or-wrong scoring

▪ MCQs cannot capture

- Calibrated judgment under uncertainty
- Hypothesis updating as new clinical data arrives
- Reasoning precision

Two physicians with identical MCQ scores can have fundamentally different clinical judgment profiles.

What We Want to Measure

The Construct

How sharply clinicians differentiate between options when data is ambiguous.

The Gap

MCQs treat clinical reasoning as binary, but it is not. Experts outperform novices when making estimates when the outcome is uncertain — a dimension traditional tests never see.

The Opportunity

Test a new item type (SCT) that quantifies diagnostic precision — with no predetermined correct answer required.

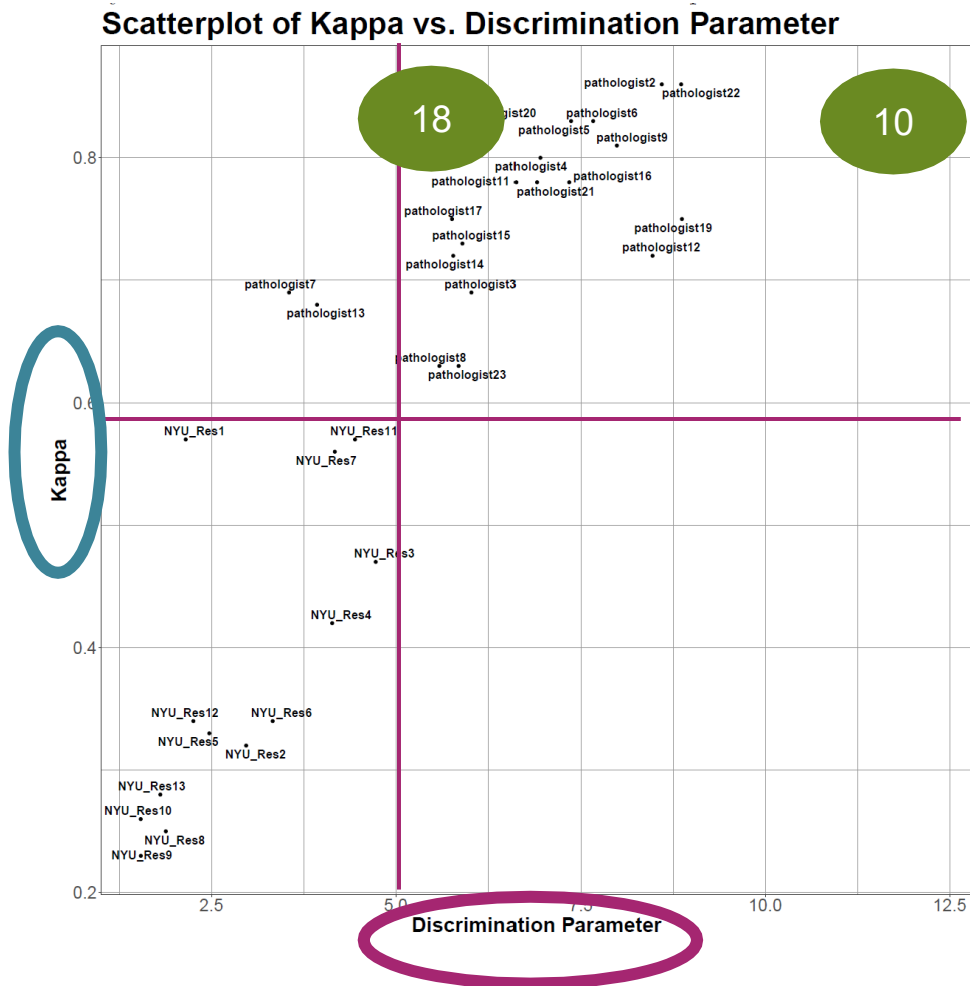
The Aspiring Examples

AN EXAMPLE FROM PATHOLOGY



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Diagnostic Precision in Pathology



1

Kappa measures agreement

How often a physician's rating matches consensus.

2

Discrimination measures precision

How sharply a physician distinguishes different levels of severity.

3

They're correlated — but not identical
Among pathologists with similar kappa (≈ 0.8), discrimination varies from 2 to 13. Pathologist 10 far outperforms Pathologist

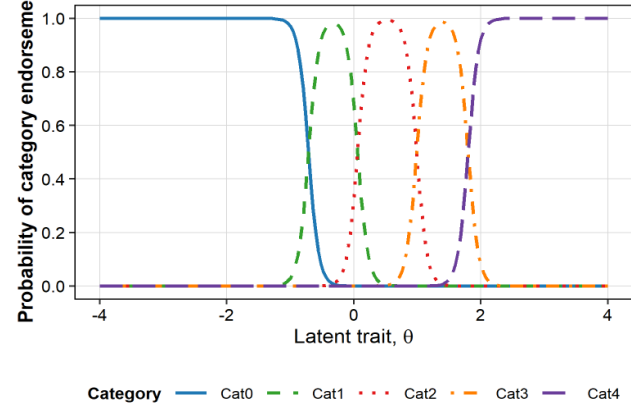
4

Why this matters

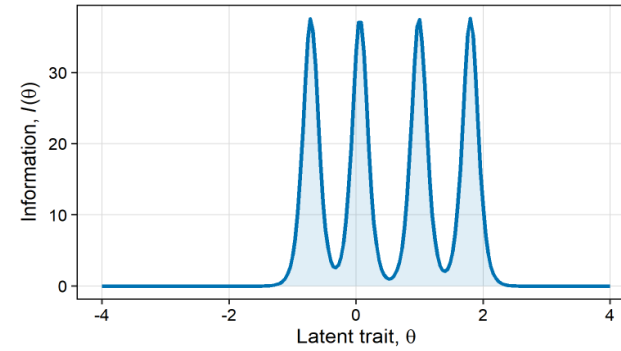
Correctness scoring misses precision. The new model captures it without needing an expert panel.

Individual Profile

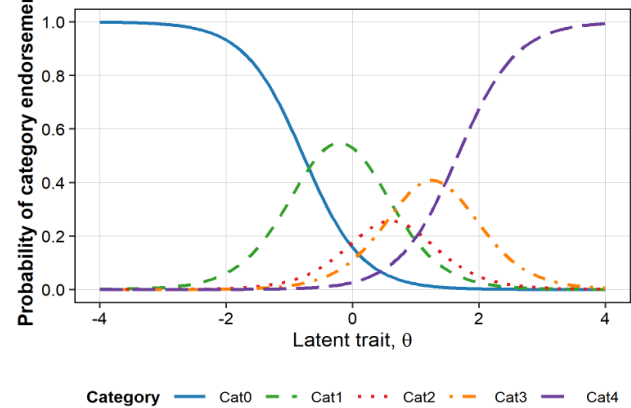
A Person Characteristic Curve – Pathologist 10 ($a = 12.27$)



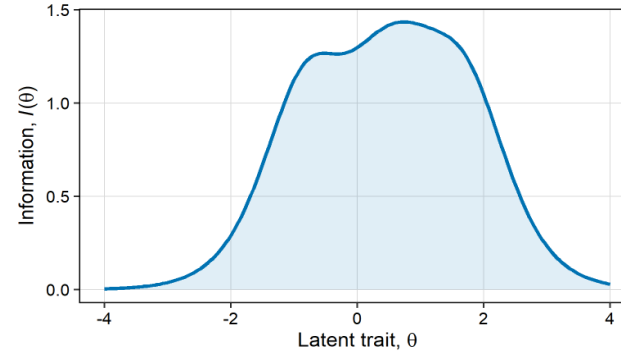
B Person Information Function – Pathologist 10 ($a = 12.27$)



C Person Characteristic Curve – NYU_Res1 ($a = 2.15$)



D Person Information Function – NYU_Res1 ($a = 2.15$)



C-Section Case

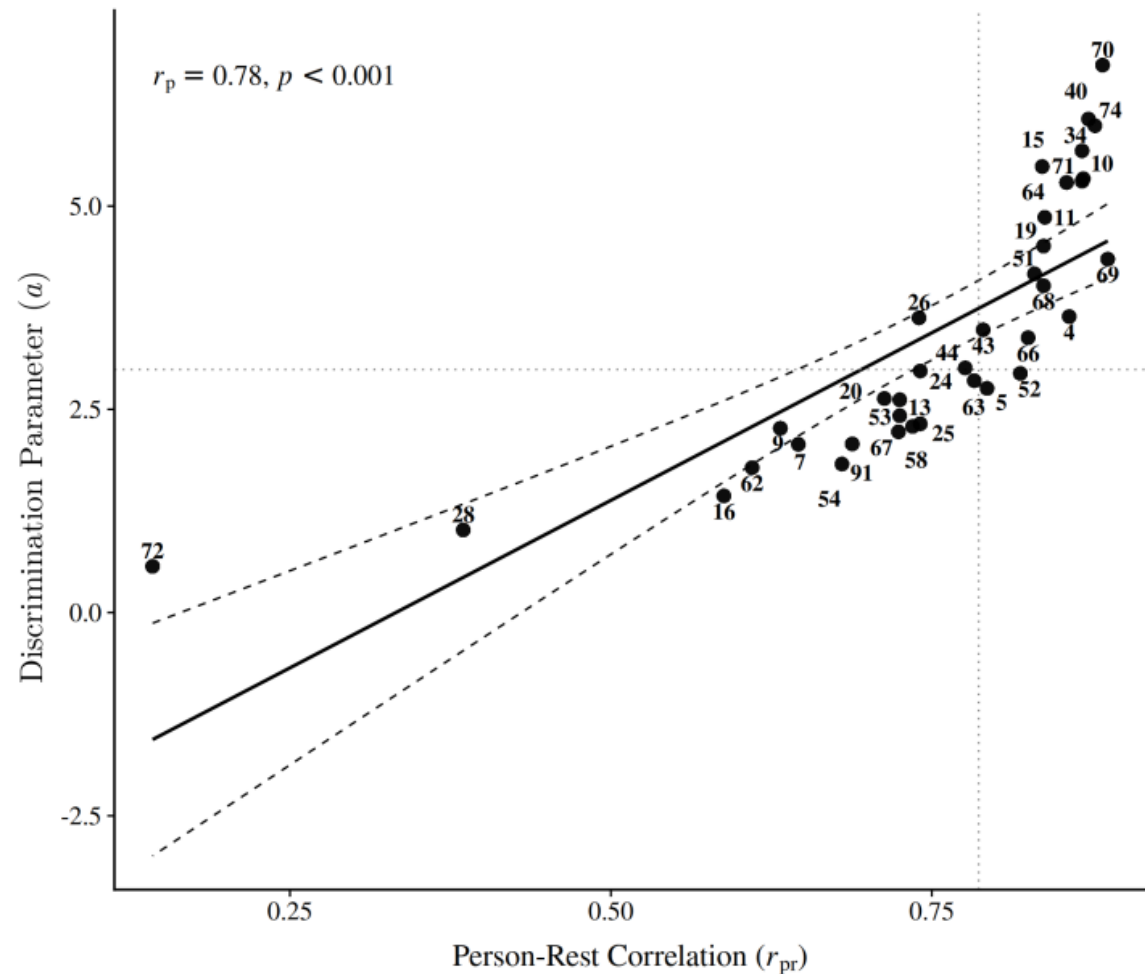
Your patient is a 29-year-old G2P1 at 39 weeks gestation. She had a previous cesarean section for breech presentation. She went into labor 12 hours ago. The patient's membranes have not ruptured, and labor is progressing steadily. She now has regular contractions, and her cervix is 5 cm dilated. The fetal heart rate tracing is Category I with moderate variability. The patient has expressed a strong preference for a vaginal birth after cesarean (VBAC) and is otherwise healthy, with no current complications. Given the situation, what would your preferred course of action be in this scenario?

- ☐ Definitely proceed with a cesarean section
- ☐ Probably proceed with a cesarean section
- ☐ Probably continue with trial of labor
- ☐ Definitely continue with trial of labor

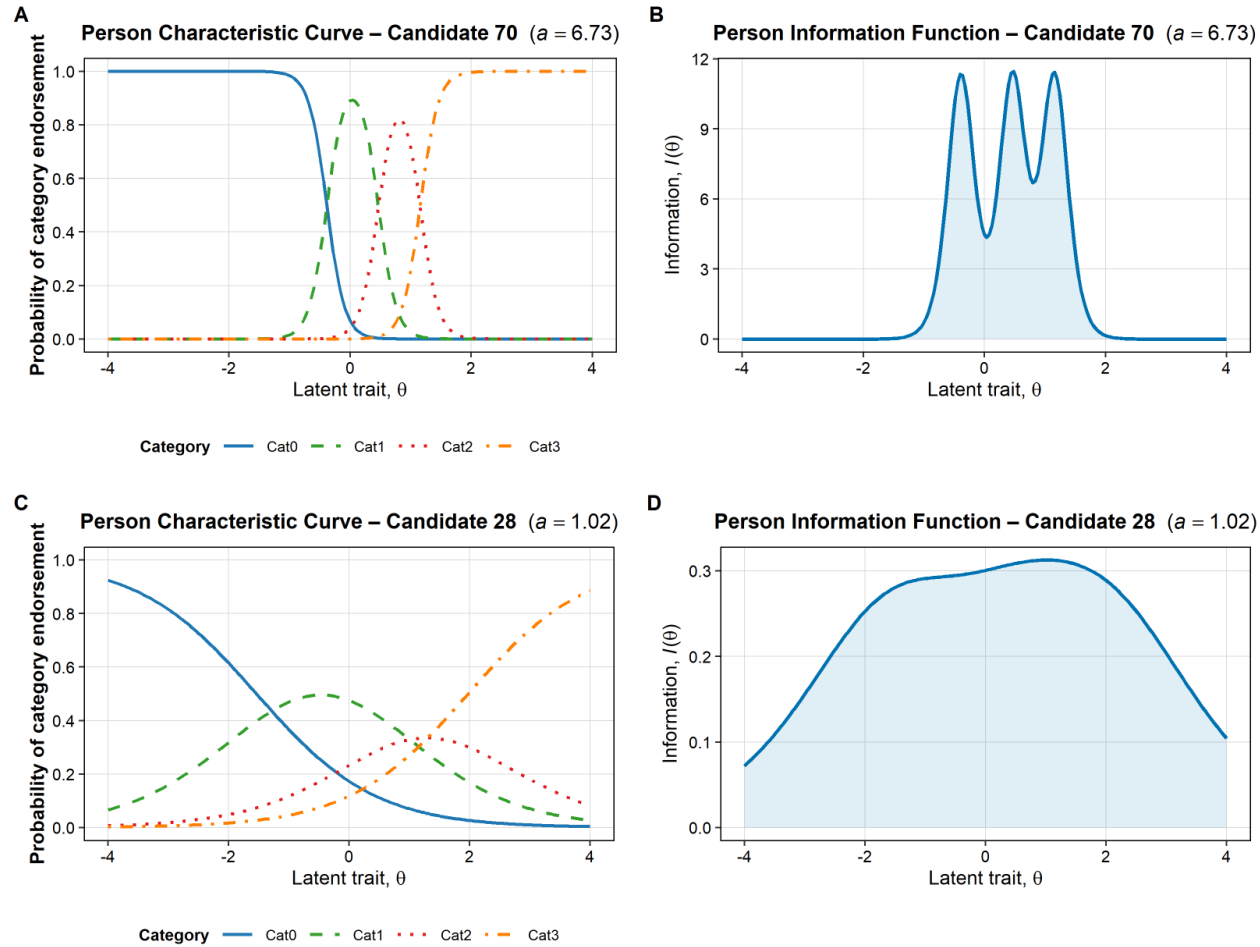
What if we don't have an expert panel?

*Decision sharpness
(Discrimination
Parameter a)*

*Consensus alignment
(Person-Rest
Correlation r_{pr})*



C-Section Individual Profile



Investigation of a New Item Type



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The Overall Construct: Assessing Diagnostic Reasoning Under Uncertainty using the EPA Framework

ENTRUSTABLE PROFESSIONAL ACTIVITY: DIAGNOSE AND MANAGE COMMON ACUTE PROBLEMS

Why EPA as the Construct?

- EPAs define real clinical tasks requiring judgment
- Diagnostic reasoning is core to clinical competency
- EPA execution requires discriminating levels of decision-making under uncertainty — precisely what we are measuring
- Applies to practicing pediatricians and trainees

Assessing and Managing Patients With
Common Behavior and Mental Health
Concerns

Not prescribe stimulant

Prescribe stimulant

Case Example: Clinical Scenario

- A 9-year-old presents with inattention, incomplete schoolwork, and fidgetiness, worsening this school year. Similar concerns were documented in prior years' report cards, predating a family move 6 months ago. Vanderbilt parent and teacher scales both meet criteria for combined-type ADHD with impairment in both settings. Anxiety and depression screens are negative. No behavioral supports have been tried.

Case Example: Options

What is the likelihood that you would prescribe a stimulant at this visit?

- A. Nearly certain (>90%) that I will prescribe
- B. Moderate (50–90%) likelihood that I will prescribe
- C. Moderate (50–90%) likelihood that I will NOT prescribe
- D. Nearly certain (>90%) that I will NOT prescribe

Case Example: Rationale

This patient meets DSM-5 symptom thresholds on both parent and teacher Vanderbilt scales with cross-setting impairment, satisfying AAP diagnostic criteria. Prior report cards document symptom onset years before the family's move, arguing against an adjustment reaction, and anxiety/depression screens are negative. For school-age children, the AAP recommends FDA-approved medication combined with parent training and school-based supports, so most physicians would prescribe a stimulant at this visit while arranging behavioral interventions — though briefly deferring medication to trial behavioral supports first is a reasonable alternative, particularly if the family is hesitant.

Candidate Feedback Survey

Post-assessment survey (1 = Strongly Disagree to 5 = Strongly Agree)

Overall Experience

1. The New Item Type felt more like real clinical decision-making than a traditional MCQ.

Relevance to Clinical Decision-Making

2. The New Item Type effectively tested my ability to update a clinical hypothesis when new information was presented.

Educational Value

3. Completing the New Item Type helped me recognize gaps in my own clinical reasoning.

Full 8-item survey administered post-assessment. Draft subject to Senior Leadership review.

Gathering Validity Evidence is Essential

Expert Group (n = 300)

- Board-certified pediatricians
- ≥ 10 years active clinical practice after training
- Testing: October 2026

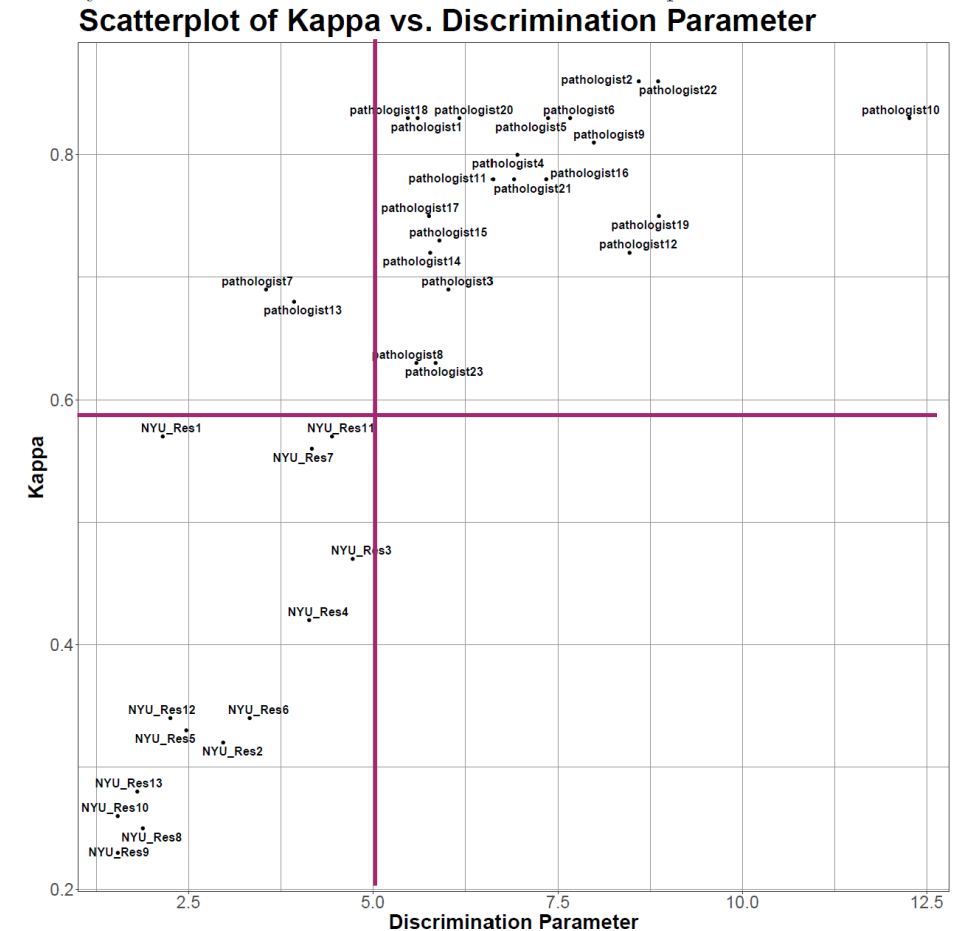
- More physicians for construct validation

Novice Group (n = 70)

- PGY-1 pediatric residents
- ≤ 6 months of clinical rotations at time of testing
- Testing: October 2025

■ Two Parallel Scoring Methods

- Same approach as pathology example from earlier
- Evaluation of 'correct' decision based on expert consensus
- Evaluation of discrimination using novel approach to psychometric analysis



Expected Results

CONVERGENT EVIDENCE FROM BOTH SCORING METHODS

Hypothesis 1

Consensus Alignment

Senior physicians score significantly higher than PGY-1 residents on kappa/Pearson-rest correlation

→ *Measure alignment of consensus*

Hypothesis 2

Flipped IRT

Senior physicians demonstrate significantly higher discrimination parameters than PGY-1 residents

→ *Validates that flipped IRT differentiates expertise without a predetermined correct answer*

Convergence of H1 + H2 validates this approach as a viable scoring alternative — without requiring an expert

panel

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Thank you!

Discussion: metric implications



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